FILTERED INDEX CATEGORIES REDUCE TO DIRECTED SYSTEMS VIA A COFINAL DIRECTED SET

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 2961-3063, following the statement at 2926-2959.

Editorial note. Conclusion (2) is reduced by the author to (1) via the C^opp dictionary with details omitted (2973-2975); the reduction is definitional (I stays filtered) and is recorded here.

Editorial note. Footnote at 2967-2969: the construction in fact produces a directed partially ordered set.

Allowed standard prerequisites (editorial).

- Colimits invariant under cofinal functors. Lemma lemma-cofinal of this chapter, invoked at proof lines 2969-2972 and 3019-3021. Transfers colimits from I to the constructed cofinal I0 and from I to I x omega via the projection Pi.
- Functorial colimit comparison transformation. Lemma lemma-functorial-colimit of this chapter, referenced in the statement at 2938-2945. The transformation theta whose invertibility is part of conclusion (1); the cofinal functor identifies both colimits.
- Cofinality of a functor. Definition definition-cofinal of this chapter, used at 3043-3061. Conditions (1) and (2) are verified for M0 at the end of the proof.

Source reference. Original definition 04E6: sources/stacks-categories/pinned-original-source.tex, source lines 1943–1943. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original definition 0030: sources/stacks-categories/pinned-original-source.tex, source lines 2824–2824. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 04E7: sources/stacks-categories/pinned-original-source.tex, source lines 1977–1977. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:categories.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 002K: sources/stacks-categories/pinned-original-source.tex, source lines 1654–1654. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 002L: sources/stacks-categories/pinned-original-source.tex, source lines 1688–1688. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{I}$ be a filtered index category. There exist a directed set I and a system $(x_i, \varphi_{ii'})$ over I in $\mathcal{I}$ with the following properties:*

(1) *For every category $\mathcal{C}$ and every diagram $M : \mathcal{I} \to \mathcal{C}$ with values in $\mathcal{C}$, denote $(M(x_i), M(\varphi_{ii'}))$ the corresponding system over I. If $\operatorname{colim}_{i\in I} M(x_i)$ exists then so does $\operatorname{colim}_{\mathcal{I}} M$ and the transformation*

$$\theta : \operatorname{colim}_{i\in I} M(x_i) \longrightarrow \operatorname{colim}_{\mathcal{I}} M$$

of Lemma 002K is an isomorphism.

(2) *For every category $\mathcal{C}$ and every diagram $M : \mathcal{I}^{opp} \to \mathcal{C}$ in $\mathcal{C}$, denote $(M(x_i), M(\varphi_{ii'}))$ the corresponding inverse system over I. If $\lim_{i\in I} M(x_i)$ exists then so does $\lim_{\mathcal{I}^{opp}} M$ and the transformation*

$$\theta : \lim_{\mathcal{I}^{opp}} M \longrightarrow \lim_{i\in I} M(x_i)$$

of Lemma 002L is an isomorphism.

Proof. As explained in the text following Definition 0030, we may view preordered sets as categories and systems as functors. Throughout the proof, we will freely shift between these two points of view. We prove the first statement by constructing a category $\mathcal{I}_0$, corresponding to a directed set[1], and a cofinal functor $M_0 : \mathcal{I}_0 \to \mathcal{I}$. Then, by Lemma 04E7, the colimit of a diagram $M : \mathcal{I} \to \mathcal{C}$ coincides with the colimit of the diagram $M \circ M_0 : \mathcal{I}_0 \to \mathcal{C}$, from which the statement follows. The second statement is dual to the first and may be proved by interpreting a limit in $\mathcal{C}$ as a colimit in $\mathcal{C}^{opp}$. We omit the details.

A category $\mathcal{F}$ is called *finitely generated* if there exists a finite set F of arrows in $\mathcal{F}$, such that each arrow in $\mathcal{F}$ may be obtained by composing arrows from F. In particular, this implies that $\mathcal{F}$ has finitely many objects. We start the proof by reducing to the case when $\mathcal{I}$ has the property that every finitely generated subcategory of $\mathcal{I}$ may be extended to a finitely generated subcategory with a unique final object.

Let ω denote the directed set of finite ordinals, which we view as a filtered category. It is easy to verify that the product category $\mathcal{I} \times \omega$ is also filtered, and the projection $\Pi : \mathcal{I} \times \omega \to \mathcal{I}$ is cofinal.

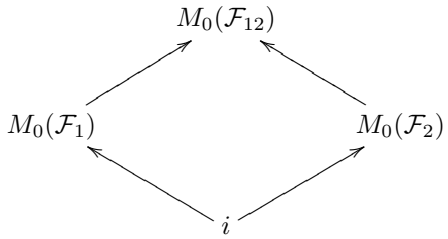
Now let $\mathcal{F}$ be any finitely generated subcategory of $\mathcal{I} \times \omega$. By using the axioms of a filtered category and a simple induction argument on a finite set of generators of $\mathcal{F}$, we may construct a cocone $(\{f_i\}, i_\infty)$ in $\mathcal{I} \times \omega$ for the diagram $\mathcal{F} \to \mathcal{I} \times \omega$. That is, a morphism $f_i : i \to i_\infty$ for every object i in $\mathcal{F}$ such that for each arrow $f : i \to i'$

[1] In fact, our construction will produce a directed partially ordered set.

in $\mathcal{F}$ we have $f_i = f_{i'} \circ f$. We can also choose i_∞ such that there are no arrows from i_∞ to an object in $\mathcal{F}$. This is possible since we may always post-compose the arrows f_i with an arrow which is the identity on the $\mathcal{I}$-component and strictly increasing on the ω-component. Now let $\mathcal{F}^+$ denote the category consisting of all objects and arrows in $\mathcal{F}$ together with the object i_∞, the identity arrow id_{i_∞} and the arrows f_i. Since there are no arrows from i_∞ in $\mathcal{F}^+$ to any object of $\mathcal{F}$, the arrow set in $\mathcal{F}^+$ is closed under composition, so $\mathcal{F}^+$ is indeed a category. By construction, it is a finitely generated subcategory of $\mathcal{I}$ which has i_∞ as unique final object. Since, by Lemma 04E7, the colimit of any diagram $M : \mathcal{I} \to \mathcal{C}$ coincides with the colimit of $M \circ \Pi$, this gives the desired reduction.

The set of all finitely generated subcategories of $\mathcal{I}$ with a unique final object is naturally ordered by inclusion. We take $\mathcal{I}_0$ to be the category corresponding to this set. We also have a functor $M_0 : \mathcal{I}_0 \to \mathcal{I}$, which takes an arrow $\mathcal{F} \subset \mathcal{F}'$ in $\mathcal{I}_0$ to the unique map from the final object of $\mathcal{F}$ to the final object of $\mathcal{F}'$. Given any two finitely generated subcategories of $\mathcal{I}$, the category generated by these two categories is also finitely generated. By our assumption on $\mathcal{I}$, it is also contained in a finitely generated subcategory of $\mathcal{I}$ with a unique final object. This shows that $\mathcal{I}_0$ is directed.

Finally, we verify that M_0 is cofinal. Since any object of $\mathcal{I}$ is the final object in the subcategory consisting of only that object and its identity arrow, the functor M_0 is surjective on objects. In particular, Condition (1) of Definition 04E6 is satisfied. Given an object i of $\mathcal{I}$, objects $\mathcal{F}_1, \mathcal{F}_2$ in $\mathcal{I}_0$ and maps $\varphi_1 : i \to M_0(\mathcal{F}_1)$ and $\varphi_2 : i \to M_0(\mathcal{F}_2)$ in $\mathcal{I}$, we can take $\mathcal{F}_{12}$ to be a finitely generated category with a unique final object containing $\mathcal{F}_1$, $\mathcal{F}_2$ and the morphisms φ_1, φ_2. The resulting diagram commutes



since it lives in the category $\mathcal{F}_{12}$ and $M_0(\mathcal{F}_{12})$ is final in this category. Hence also Condition (2) is satisfied, which concludes the proof. □